Life Cycle Assessment of Duckweed Biorefineries for Drying and Protein Production

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Introduction

- Duckweed (commonly known as *Lemnaceae*) are a promising alternative protein source and have been approved as a safe, novel food (Ujong et al., 2025).
- Duckweed processing includes converting fresh or dried duckweed biomass into protein ingredients, particularly through different drying and extraction technologies, however environmental impacts of processing stages remain unclear (Maag et al., 2025).

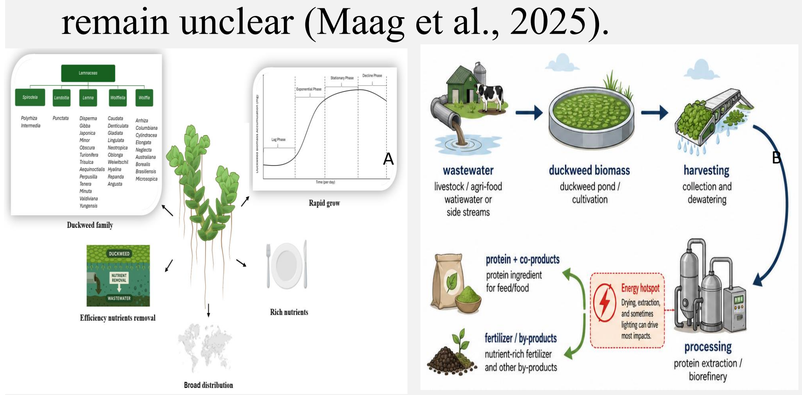


Fig.1: *(A) The advantages of duckweed as a safe, novel potential protein source. (B) Current duckweed biorefinery system*

Method

Three system boundaries were evaluated: duckweed drying, protein extraction from dried duckweed, and protein extraction from frozen-thawed duckweed and LCA models were designed based on Ecoinvent database with possible validation points.

A "gate-to-gate" life cycle assessment approach was adopted, in which the duckweed protein extraction process was divided into three distinct systems: (i) drying, (ii) protein extraction from dried duckweed, and (iii) protein extraction from frozen-thawed duckweed.

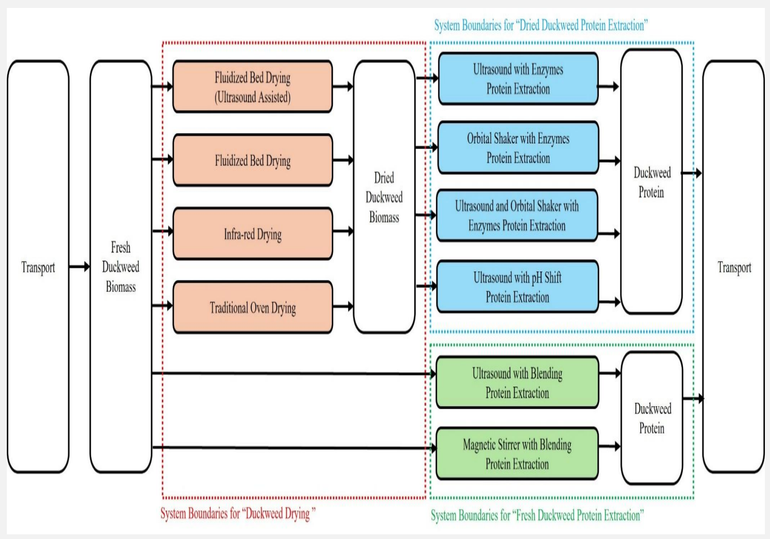


Fig.2: *System boundaries for duckweed drying dried duckweed protein extraction and frozen-thawed fresh duckweed protein extraction*

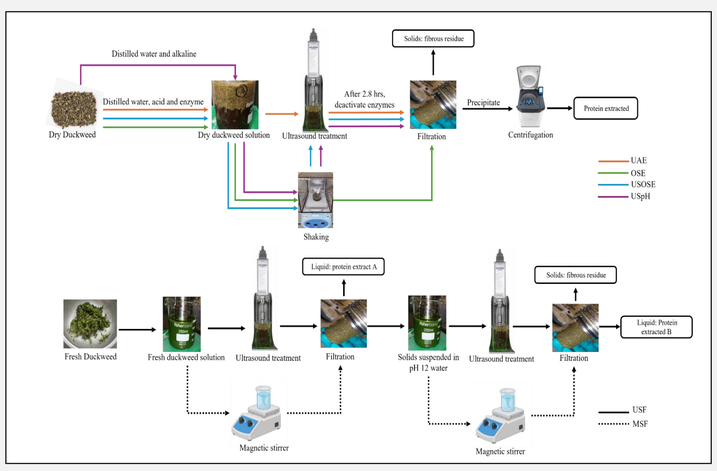


Fig.2: *Dried and frozen-thawed duckweed extraction workflow*

Scenario analysis and Monte Carlo simulation was used as uncertainty analysis. All results were calculated by SimaPro 10.3.0.1.

Results

LCI results

USE achieved the highest performance ratio among the dried-duckweed extraction routes, whereas OSE exhibited the lowest. For frozen-thawed duckweed protein extraction, USF showed the highest performance ratio.

	USE	OSE	USpH	USOSE	USF	MSF
Recover yield (%)	61.7 ± 2.2	57.1 ± 1.0	35.9 ± 3.1	36.8 ± 1.8	47.9 ± 1.0	50.6 ± 0.2
Power (kWh)	0.04±0.01	1.31±0.01	0.95±0.01	0.94±0.05	0.03±0.01	0.11±0.01

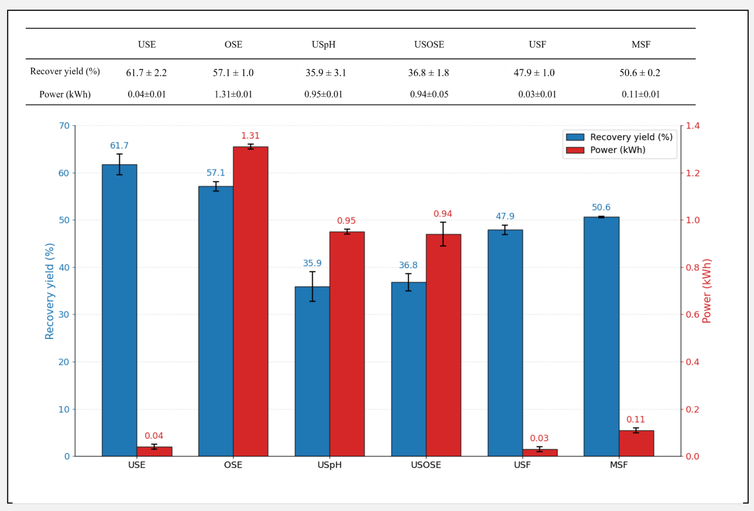


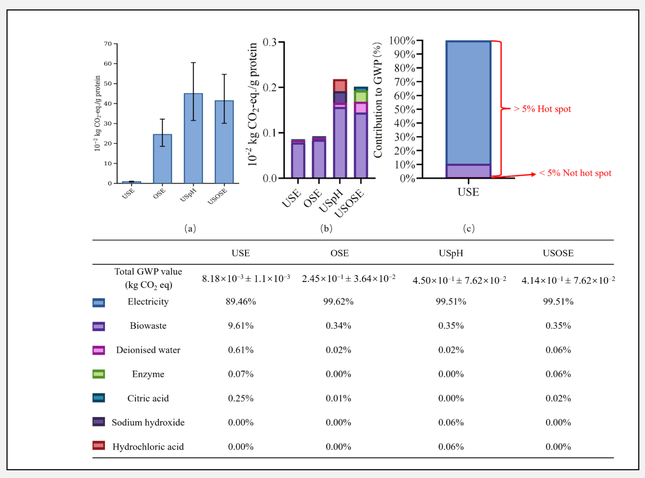
Fig.4: *Recover yield and electricity power consumption of each extraction process*

LCA results of drying

The FBD (baseline model) had the lowest global warming potential (GWP), averaging 6.0×10^{-2} kg CO_2 eq, followed by the USFBD. The IR, as an emerging drying method, did not show superior environmental impact potential in this experiment, differing from the conventional oven process by only $3.0\times10^{-}2$ kg CO_2 eq.

LCA results of dried duckweed protein extraction

USE (baseline model) had the lowest GWP of $8.18\times10^{-3} \pm 1.1\times10^{-3}$ kg CO_2 eq, followed by OSE with a GWP value of $2.45\times10^{-1} \pm 3.64\times10^{-2}$ kg CO_2 eq. USOSE and USpH ranked third ($4.50\times10^{-1} \pm 7.62\times10^{-2}$ kg CO_2 eq) and fourth ($4.14\times10^{-1} \pm 7.62\times10^{-2}$ kg CO_2 eq) respectively.

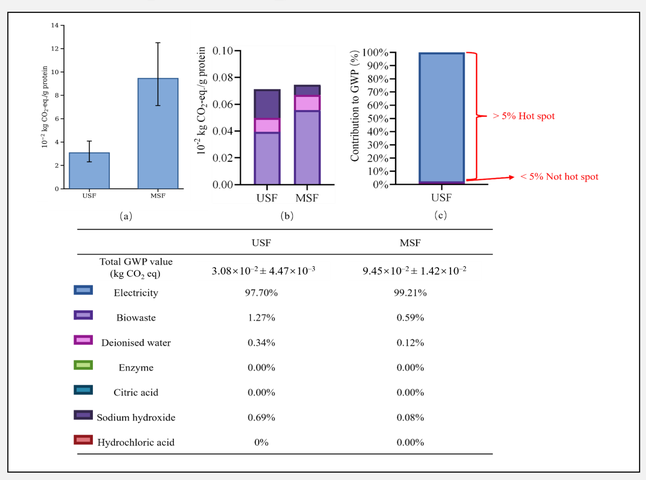


	USE	OSE	USpH	USOSE
Total GWP value (kg CO_2 eq)	$8.18\times10^{-3}\pm1.1\times10^{-3}$	$2.45\times10^{-1}\pm3.64\times10^{-2}$	$4.50\times10^{-1}\pm7.62\times10^{-2}$	$4.14\times10^{-1}\pm7.62\times10^{-2}$
Electricity	89.46%	99.62%	99.51%	99.51%
Biowaste	9.61%	0.34%	0.35%	0.35%
Deionised water	0.61%	0.02%	0.02%	0.06%
Enzyme	0.07%	0.00%	0.00%	0.06%
Citric acid	0.25%	0.01%	0.00%	0.02%
Sodium hydroxide	0.00%	0.00%	0.06%	0.00%
Hydrochloric acid	0.00%	0.00%	0.06%	0.00%

Fig. 5: *GWP of dried duckweed protein extraction: (a) Contributions of different input categories for GWP; (b) Contributions of different input categories for GWP without electricity consumption; (C) Hotspot analysis for USE*

LCA results of frozen-thawed duckweed protein extraction

USF (baseline model) had a substantially lower GWP ($3.08\times10^{-2} \pm 4.47\times10^{-3}$ kg CO_2 eq), approximately one-third of MSF ($9.45\times10^{-2} \pm 1.42\times10^{-2}$ kg CO_2 eq).



	USF	MSF
Total GWP value (kg CO_2 eq)	$3.08\times10^{-2}\pm4.47\times10^{-3}$	$9.45\times10^{-2}\pm1.42\times10^{-2}$
Electricity	97.70%	99.21%
Biowaste	1.27%	0.59%
Deionised water	0.34%	0.12%
Enzyme	0.00%	0.00%
Citric acid	0.00%	0.00%
Sodium hydroxide	0.69%	0.08%
Hydrochloric acid	0%	0.00%

Fig. 6: *GWP of frozen-thawed duckweed protein extraction: (a) Contributions of different input categories for GWP; (b) Contributions of different input categories for GWP without electricity consumption; (C) Hotspot analysis for USF*

Scenarios and sensitive analysis

Compared to the base line model, all three scenarios reduced environmental burdens (↓3 - 98%), identifying electricity sourcing (↓97 - 98%), residual-biomass management (↓3 - 11%), and process scale (↓63 - 71%) as key optimisation priorities.

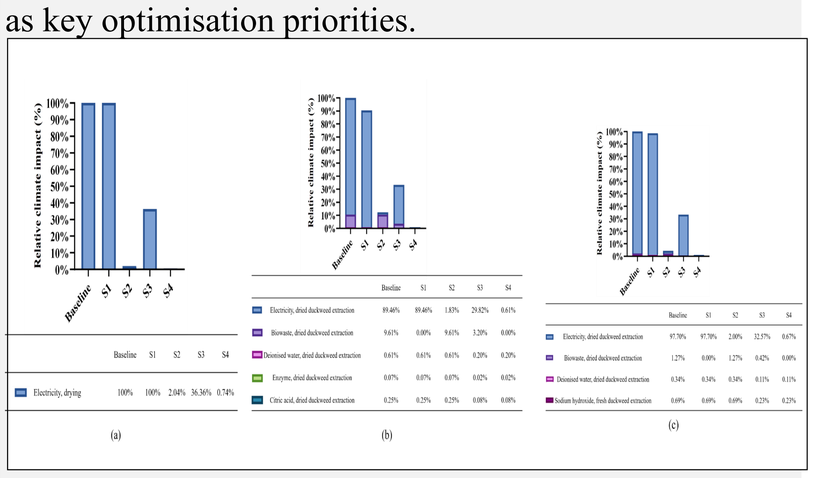


Fig. 7: *(a) Relative climate impacts under different scenarios for electricity consumption in duckweed drying (b) Relative climate impacts under different scenarios for electricity consumption in dried duckweed protein extraction (c) Relative climate impacts under different scenarios for electricity consumption in fresh duckweed protein extraction*

Discussion

- Electricity consumption and biowaste treatment were identified as common environmental hotspots across the three systems. Traditional wet protein extraction methods consume substantial energy because they involve multiple stages of water addition and removal.
- The more rigid structure of plant cells requires extra disruption and separation steps, which generally consume more electricity. Adopting Irish electricity mix data as the electricity source inevitably introduces the environmental impacts associated with conventional power generation into the LCA system.
- Laboratory-scale energy hotspots and should not be interpreted as definitive evidence of the inherent environmental disadvantages of the technologies.
- Pilot-scale measurements and an integrated assessment covering cultivation, drying, extraction, downstream processing, and residue utilisation are required for environmental performance of complete duckweed protein-production pathways.

References

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